



A reaction mechanism for gasoline surrogate fuels for large polycyclic aromatic hydrocarbons

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ABSTRACT

This work aims to develop a reaction mechanism for gasoline surrogate fuels (n-heptane, iso-octane and toluene) with an emphasis on the formation of large polycyclic aromatic hydrocarbons (PAHs). Starting from an existing base mechanism for gasoline surrogate fuels with the largest chemical species being pyrene ($C_{16}H_{10}$), this new mechanism is generated by adding PAH sub-mechanisms to account for the formation and growth of PAHs up to coronene ($C_{24}H_{12}$). The density functional theory (DFT) and the transition state theory (TST) have been adopted to evaluate the rate constants for several PAH reactions. The mechanism is validated in the premixed laminar flames of n-heptane, iso-octane, benzene and ethylene. The characteristics of PAH formation in the counterflow diffusion flames of iso-octane/toluene and n-heptane/toluene mixtures have also been tested for both the soot formation and soot formation/oxidation flame conditions. The predictions of the concentrations of large PAHs in the premixed flames having available experimental data are significantly improved with the new mechanism as compared to the base mechanism. The major pathways for the formation of large PAHs are identified. The test of the counterflow diffusion flames successfully predicts the PAH behavior exhibiting a synergistic effect observed experimentally for the mixture fuels, irrespective of the type of flame (soot formation flame or soot formation/oxidation flame). The reactions that lead to this synergistic effect in PAH formation are identified through the rate-of-production analysis.

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1. Introduction

Gasoline fuel consists of several hydrocarbon groups including straight and branched chain aliphatics together with aromatics. Among them, n-heptane (nC_7H_{16}), iso-octane (iC_8H_{18}) and toluene ($C_6H_5CH_3$) are recognized as the primary constituents in representing gasoline surrogate fuels [1]. Recently, several theoretical [1–6] and experimental studies [7–14] have been conducted on one or more of these components to determine their burning velocities, ignition delay times, and the concentrations of molecular species formed during their burning in various combustion environments. Such studies can be useful in gasoline blending and fuel formulation concerning the compositions of different fuel components.

Studies on the formation of pollutants such as polycyclic aromatic hydrocarbons (PAHs) and soot from these fuels, however, are rather limited [3,15–17]. The presence of aromatics in transportation fuels such as gasoline promotes the formation of PAHs and soot particles, which are known to cause adverse health problems [18]. While toluene could act as octane booster in suppressing knock.

Recently, direct injection spark-ignition engines (DISI) are being extensively developed for fuel efficiency and emission control, which may lead to increased formation of soot [3]. This can pose a problem in the future due to potential emission regulation on the size and number of particulate matter in engine exhaust. Therefore, accurate prediction of the concentrations of PAHs and soot formed in combustion devices is required.

To predict the concentrations of small hydrocarbons as well as PAHs during the combustion of gasoline surrogate fuels, several kinetic mechanisms are available in the literatures [1,2,5,16,19], which have been shown to effectively predict the experimental observations such as chemical species profiles (e.g. C_2H_2 and C_6H_6), laminar burning velocities and ignition delay times. However, only few of them have been validated against the profiles of large polycyclic aromatic hydrocarbons (PAHs) [16].

In our recent work on PAH and soot formation in the counterflow flames of gasoline surrogate fuels of n-heptane/toluene and iso-octane/toluene mixtures [3], the synergistic effect on PAH formation has been observed experimentally, demonstrating enhanced formation of PAHs for mixture fuels as compared to respective pure fuels [20–23]. Two recently developed mechanisms for high temperature combustion applications [1,2] were tested to predict the synergistic behavior. It was found that the

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predictions based on these mechanisms either could not exhibit the synergistic effect for PAHs larger than benzene, or appreciably over-predict the synergistic effect for a certain type of counterflow flame.

This implies that the sub-mechanisms for the growth of PAHs in these mechanisms require improvement. Several studies have been published in the last decade on the role of large PAHs in the formation of soot particles through their physical coagulation ([24] and refs. therein). Clearly, without an accurate estimate of the amount of PAHs formed in a flame zone, a reliable prediction of soot observables such as soot volume fraction, number density and particle size distribution from soot models cannot be obtained. This highlights the need to investigate the pathways for the growth of PAHs during the combustion of large hydrocarbons, and include them in reaction mechanisms.

The objective of the present study is to extend the reaction mechanism for gasoline surrogate fuels by including and modifying sub-mechanisms for PAH growth to improve the model prediction of PAH concentrations in various flame environments. For this, the rate constants for several PAH reactions have been evaluated from the first principles using the density functional theory (DFT) and transition state theory (TST). New reaction pathways to account for the growth of PAHs up to coronene ($C_{24}H_{12}$) have been explored. The reactions involved in them are added to the mechanism, and wherever required, their rate constants are updated with recent estimates. Thereafter, the new reaction mechanism is validated in several premixed laminar flames and counterflow diffusion flames using available quantitative and qualitative information on PAHs, followed by a detailed analysis of the reaction pathways of the new mechanism.

2. Kinetic model development and simulation

2.1. Quantum calculations

Quantum chemical calculations on hydrocarbons are generally conducted to predict their optimized structures and thermochemical properties along with other useful information such as vibrational frequencies, moments of inertia and reaction energetics. Several methods are proposed in literatures for quantum calculations such as *ab initio* methods (for example, Hartree Fock theory, Coupled Cluster (CC) theory), semi-empirical methods (for example, Austin Model 1) and density functional theory (DFT), which are detailed in [18] and references therein. Coupled-cluster singles, doubles and perturbative triples, CCSD(T) [25] is considered to be very accurate for quantum calculations, but are computationally demanding. With the present computing resources, such a high level calculation can be adopted for small molecules in a reasonable computational time. In this work, quantum calculations were required to be conducted on PAH molecules, which are sufficiently large to make quantum calculations on them computationally expensive. For such molecules, DFT is widely adopted, as it is computationally more affordable than higher levels of theory while providing reasonable accuracy in the results. Therefore, DFT has been adopted for the quantum calculations in this study.

For each PAH reaction studied in this work, the molecular structures of the chemical species and the transition state were optimized with B3LYP hybrid functional [26] and 6-311++G(d,p) basis set [27] using Gaussian 09 [28]. For stable chemical species, the structures were optimized with different spin multiplicities to determine the one with minimum energy, reasonable geometry and negligible spin contamination [29]. Vibrational frequencies were also calculated, and it was ensured that the stable species do not have any imaginary frequencies and the transition states have only one imaginary frequency. The vibrational mode of the

imaginary frequency of all the transition states was animated to confirm that the motion correctly connects the corresponding reactants and products. B3LYP/6-311++G(d,p) level of theory has already been employed successfully in studying PAH reactions in our previous works [29,30], where it has been validated extensively by comparing the computed results with experimental observations.

To evaluate the rate constants of PAH reactions, the transition state theory (TST) was used along with the Eckart correction factor [31] to account for the quantum tunneling. Otherwise, the rate constants for elementary reactions can be under-estimated [29]. Vibrational frequencies, moment of inertia, mass and spin multiplicities of reactants, products and the transition state, required for chemical kinetics calculations by TheRate software [31], were obtained from DFT calculations. A modified Arrhenius expression ($k(T) = AT^n \exp(-E/RT)$) was fitted to the calculated rate constants in the temperature range of 300–3000 K to obtain the kinetic parameters of the frequency factor A , the temperature exponent n and the activation energy E .

2.2. Flame simulations

Soot and PAH characteristics in premixed flames could be different from those in diffusion flames. Thus, for a mechanism to be applicable in a practical combustion environment, it should be able to predict the experimental observations for all types of flames. Therefore, premixed flames as well as diffusion flames have been simulated for mechanism validation.

Two sets of experimental data have been tested in this study – one for premixed laminar flames and the other for counterflow diffusion flames. The purpose of simulating the premixed flames was to assess the quantitative prediction capability of the reaction mechanism using the existing experimental data for single component fuels such as n-heptane, iso-octane and benzene along with ethylene [32], which has been most extensively investigated for soot [32–36] and PAH [32] studies. The purpose of simulating counterflow diffusion flames of the binary fuels of n-heptane/toluene and iso-octane/toluene was to assess the prediction capability for mixture fuels. The accurate prediction for mixture fuels, especially involving surrogates of real fuels such as gasoline, is important due to possible crosslinking between fuel constituents [3,21].

Table 1 provides the details of premixed laminar flames simulated in this work for the mechanism validation. For the premixed laminar flames, experimentally observed temperature profiles were used for calculations, as they were available. The flames of n-heptane and iso-octane, PF1a and PF1b, have been simulated to compare the experimentally observed profiles of the reactants and small chemical species up to benzene (A1) with the computed ones. This is followed by the comparison of the computed and the observed profiles of PAHs up to pyrene (A4) in n-heptane flames (PF2a and 2b). Due to the absence of the information on PAHs larger than A4 in the flames of gasoline surrogate fuels, ethylene (PF3) and benzene (PF4) flames having available experimental data on PAHs with up to six aromatic rings are simulated to assess the capability of the new mechanism.

The counterflow diffusion flames of iso-octane – toluene and n-heptane – toluene mixtures experimentally observed in [3] were simulated. For the counterflow flames, the energy equation was solved to obtain the temperature variation, as the experimental profiles were not available. All the flames were simulated using CHEMKIN PRO [37].

In counterflow flames, sooting characteristics can be varied significantly depending on relative flame position to the stagnation plane [38]. If the flame is on the fuel side, since soot particles are generated on the fuel side of the flame, they will migrate towards

Table 1

Test conditions of premixed laminar flames at atmospheric pressure studied in this work.

Flame	Fuel	Composition (mole %)			Cold gas velocity (cm/s)	Ref.
		Fuel	O ₂	N ₂ or Ar		
PF1a	nC ₇ H ₁₆	3.98	23.01	73.01 (N ₂)	4.98	[2,8]
PF1b	iC ₈ H ₁₈	4.23	27.80	67.97 (N ₂)	4.18	[2,8]
PF2a	nC ₇ H ₁₆	5.50	28.79	65.71 (Ar)	5.17	[7]
PF2b	nC ₇ H ₁₆	5.33	29.70	64.97 (Ar)	5.25	[7]
PF3	C ₂ H ₄	21.3	20.9	57.8 (Ar)	6.42	[32]
PF4	C ₆ H ₆	4.8	20.0	75.2 (N ₂)	4.0	[57]

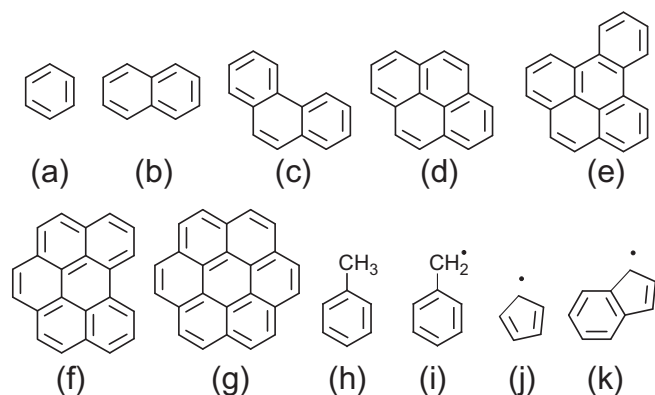


Fig. 1. Structures of important species considered. (a) benzene (A1). (b) naphthalene (A2). (c) phenanthrene (A3). (d) pyrene (A4). (e) benzo[e]pyrene (A5). (f) benzo[ghi]perylene (A6). (g) coronene (A7). (h) toluene (C₆H₅CH₃). (i) benzyl radical (C₆H₅CH₂). (j) cyclopentadienyl radical (C₅H₅) (k) indenyl radical (C₉H₇).

the flame and subsequently get oxidized by OH or O₂. Such flames are called soot formation/oxidation (SFO) flames. When the flame is on the oxidizer side, soot particles will be convected away from the flame, and will not be oxidized. Such flames are named as the soot formation (SF) flames.

In total, 44 counterflow flames consisting of 22 SF and SFO flames each were simulated, among them eleven flames each for iso-octane/toluene and n-heptane/toluene mixtures. For all the counterflow flames, the oxidizer flow velocity (O₂ + N₂) was 21 cm/s and the fuel flow velocity was 28 cm/s to conform with the experiments. The mole fractions of fuel and oxygen were 0.3 and 0.21 for SF flames and 0.07 and 0.9 for SFO flames, respectively, at the nozzle exit which are balanced with N₂. The toluene ratio in the fuel mixture, R_T , was varied between 0 and 1 uniformly in the SF and the SFO flames. Details about the experimental conditions of the flames can be obtained from [3].

2.3. Mechanism analysis

An analysis of the developed reaction mechanism has been carried out to determine the reactions and gas-phase species responsible for the synergistic behavior of four PAHs, A1, A2, A3 and A4 (see Fig. 1 for the structures of the chemical species, which have been frequently referred to in this paper) in the soot formation/oxidation flames of iso-octane/toluene mixtures. The procedure is briefly described here.

The rate-of-production analysis [37] is conducted using CHEM-KIN PRO to determine the reactions that mainly contribute to the formation of these PAHs at different temperatures and for different R_T values. A list is generated for all the chemical species involved in those reactions and their normalized maximum mole fractions for different values of R_T are calculated. Those reactions in the form of $A + B \rightarrow Ai + C$ involving two gas-phase species (A and B) to form a PAH Ai ($i \in \{1, 2, 3, 4\}$), are chosen, in which, either the normalized

maximum mole fractions of both the reactants show opposite trends (that is, one increasing and the other decreasing with R_T), or one of the two reactants shows the synergistic behavior. The reactions in the form of $D \rightarrow Ai + E$ are chosen only when D shows the synergistic behavior. All the chosen reactions are then tested to determine those responsible for the synergistic effect of the PAHs. The tests are conducted for each PAH molecule (A1–A4) by firstly removing all the chosen reactions for a PAH molecule from the mechanism (which leads to the loss of synergistic effect), and then selectively inserting back the reactions in the mechanism until the synergistic effect for the PAH is obtained. Since more than one reaction can be responsible for the synergistic effect of a PAH molecule, the above step is repeated several times until the list of the chosen reactions is exhausted.

3. Results and discussion

3.1. Mechanism development

As mentioned previously, two recent mechanisms for gasoline surrogate fuels, Mechanism 1 [1] and Mechanism 2 [2] were not satisfactory in predicting the experimentally observed synergistic effect for PAHs in the counterflow flames [3]. This section details the steps that were taken to develop PAH growth sub-mechanisms present in the reaction mechanism to improve the model-predicted concentrations of PAHs.

Mechanism 2 [2], which has been validated in several flame environments through the comparison of species profiles for small hydrocarbons up to benzene, was used as the base mechanism. The reason behind that was its ability to predict the synergistic effect of PAHs in the soot formation/oxidation flames of iso-octane/toluene mixtures (except for A2) and in the soot formation flames of iso-octane/toluene and n-heptane/toluene mixtures (although the effect was highly over-predicted for some PAHs) [3]. The thermodynamic data for all the species present in the base mechanism were taken from [2] without any alterations.

This mechanism had pyrene (A4) as the largest PAH molecule and the hydrogen-abstraction-C₂H₂/C₄H₄-addition (HACA) mechanism was mainly adopted for PAH growth. It will be shown later that, with this mechanism, the concentrations of the large PAHs (in particular, pyrene) in a n-heptane flame were considerably under-predicted. Consequently, several new PAH reactions were added to this mechanism to account for the formation and growth of PAHs up to coronene (C₂₄H₁₂). The rate constants for some of the PAH reactions present in the base mechanism were updated with their recent estimates from the literatures (for example, from [1,16,29,39,40]). The purpose of extending the mechanism up to coronene was to make the new mechanism suitable for its use in soot modeling, since PAHs larger than pyrene are thought to inception soot particles [24]. The reactions involving chemical species with less than five carbon atoms were not altered in the base mechanism, which were mainly obtained from the ABF mechanism [41] and the mechanism developed by Kyne et al. [42].

Table 2

Benzene reactions added based on [47]. Rate constants in the form of $AT^n \exp(-E/RT)$ with the units of cal, K, mol, cm and s, and “.” representing radical sites.

Reaction	A	n	E	Ref.
$C_5H_5 + CH_3 \rightarrow C_5H_5CH_3$	1.50×10^{15}	-1.527	-13,500	[47]
$C_5H_5CH_3 \rightarrow C_5H_5 + CH_3$	6.59×10^{59}	-13.606	82,000	[47]
$C_5H_5CH_3 \rightleftharpoons C_5H_4CH_2 + H_2$	1.46×10^{48}	-12.326	80,000	[47]
$C_5H_5 + CH_3 \rightleftharpoons C_5H_4CH_2 + H_2$	1.50×10^{11}	-0.3452	16,631	[47]
$C_5H_4CH_2 \rightleftharpoons A1$	1.45×10^{45}	-8.900	97002.87	[1]
$C_5H_4CH_2 \rightleftharpoons A1 + H$	2.24×10^{68}	-14.650	142576.48	[1]

Oxygenated PAH molecules are readily formed in combustion processes [43,44], which can, in turn, form hydrocarbons with odd number of carbon atoms through the loss of CO molecule. Nine such species and 59 reaction involving them [1] were added to the base mechanism. In toluene-rich flames, the base mechanism was found to appreciably under-predict the concentrations of large PAHs (mainly A3 and A4). Therefore, twelve species and 72 reactions accounting for toluene oxidation and decomposition [1,16,45] were added. For these newly added chemical species, the thermodynamic data and the transport data were obtained from the Supplementary materials of [1,16,45]. The DFT calculations were conducted on those PAH reactions for which a discrepancy in their rate constants was found in the literatures, or an update in their rates was required. The rate constants for some of the PAH reactions were updated with their recent estimates from our previous studies [29,40]. The reactions, which were added specifically for the formation of the PAHs important for this work (i.e., A1–A7), are detailed in Sections 3.1.1, 3.1.2, 3.1.3, 3.1.4, 3.1.5.

The sub-mechanisms for the decomposition of n-heptane, iso-octane and n-decane, present in the base mechanism, were not altered. Note that, in this work, none of the rate constants have been arbitrarily changed or intuitively estimated to match the simulation results with the experimental observations. In total, the base mechanism consisted of 154 chemical species and 1404 reactions (counting forward and backward reactions separately). After the addition of new PAH reactions, the new mechanism consisted of 226 chemical species and 2121 reactions (counting forward and backward reactions separately). Several selected reactions are listed in Table 2 for benzene formation and in Table 3 for A2–A7.

3.1.1. Benzene (A1) reactions

It is mentioned in [46] that benzene can form through various reaction channels depending upon flame conditions. The common pathways considered in most of the mechanisms involve propargyl (C_3H_3) recombination and the addition of C_2H_x on C_4H_y molecules. In [47], a new pathway for the formation of benzene was studied through the addition of CH_3 on cyclopentadienyl (C_5H_5) radicals. The reactions involved in this pathway were thought to be important due to the formation of appreciable amount of C_5H_5 during the combustion of aromatic compounds, and thus, were included in the mechanism. The newly added reactions are listed in Table 2. The thermodynamic data for the species, $C_5H_5CH_3$ and $C_5H_4CH_2$ were obtained from [1,48], respectively.

3.1.2. Naphthalene (A2) reactions

One of the important pathways for the formation of naphthalene is through the self-combination of cyclopentadienyl radicals (C_5H_5) [49]. In the base mechanism, the chemical species with five-member rings such as C_5H_5 and cyclopentadiene (C_5H_6) were not present. Therefore, 57 reactions involving these species and their thermodynamic data were added from [1,45,50,51]. Similarly, indene (C_9H_8) and its radicals can play significant roles in the

formation and growth of PAHs. Such species with their reactions and thermodynamic data were also added from [1,16,48]. The rate constants for the H abstraction taking place on the chemical species such as A2, C_5H_6 and C_9H_8 by H, CH_3 , and C_3H_3 and H addition on their radicals by H_2 , CH_4 and C_3H_4 were evaluated in this work (see Table 3), as such reactions were found to be important in predicting the synergistic effect for A2 (shown later).

New rates for the reactions involved in the formation of A2 from A1 through HACA mechanism were also evaluated (see Table 3). It will be shown in Section 3.3.2 that A2 formation through this pathway with the updated rates leads to the successful prediction of its synergistic effect. The detailed pathways for the formation of A2 from benzyl radical, $C_6H_5CH_2$ (formed from toluene) and C_3 species were also included from [45].

3.1.3. Phenanthrene (A3) reactions

In the base mechanism, the reactions forming A3 mainly involved the addition of benzene or its radical on an alkylated benzene or its radical, the addition of C_2H_2 on a biphenyl radical, and the growth of A2 through the HACA mechanism. The rates for some of the elementary reactions involved in these pathways were updated by either conducting DFT calculations on them or by using their recent estimates from the literature, as shown in Table 3. Two pathways for the reaction between $A1C_2H$ and $A1$ (reactions N21 and N22) were considered in this work, and are shown in Fig. 2. The rate constants for the reactions involving the addition of $A1$ on $A1C_2H$ in the two pathways were assumed to be the rate constants for the overall pathways, and are provided in Table 3.

The pathways for the formation of A3 through the radicals formed from toluene decomposition were not present in the base mechanism. Two such pathways, one involving the self-addition of benzyl radicals [45] and the other involving the addition of an indenyl radical (C_9H_7) on a cyclopentadienyl radical (C_5H_5) [1] were included in the mechanism. Other formation pathways for A3 include the addition of C_2 and C_3 species on alkylated naphthalenes or their radicals.

3.1.4. Pyrene (A4) reactions

Similar to A3, the new formation pathways for A4 that were added to the mechanism involve chemical species generated from toluene decomposition such as $C_6H_5CH_2$ and the species with odd number of carbon atoms such as C_5H_5 and C_9H_7 , as shown in Table 3. Note that many pathways for the formation of A3 and A4 involve one-step reactions, whose rates have been obtained from the literatures [1,52,53]. These pathways need to be studied in detail in the future to determine the elementary steps involved in them, and obtain more accurate estimates of their rate constants.

3.1.5. Benzo[e]pyrene (A5), benzo[ghi]perylene (A6) and coronene (A7) reactions

Twenty-six species and 77 reactions were added in the base mechanism to account for the formation of PAHs and their radicals with molecular mass higher than pyrene and up to coronene. Most of the reactions and the rates for PAH growth beyond pyrene were taken from [1,52–54]. These literature-based reactions mainly involved PAH growth through the addition of C_2H_2 on PAH radicals. In the present work, several reactions for the growth of large PAHs by C_4H_4 have been added. Furthermore, new H-abstraction reactions for all the PAHs beyond pyrene by the reaction with CH_3 and C_3H_3 have been included in this study [55]. For all the H-abstraction reactions (including abstraction by H atoms), the rate constants evaluated for A2 or A3 in this work have been used depending on the type of reactive sites present on the larger PAHs [39], as DFT calculations on large PAHs are computationally very demanding. It is mentioned in [39] (and, can also be seen in [56]) that the rate constant for a reaction taking place on a large PAH

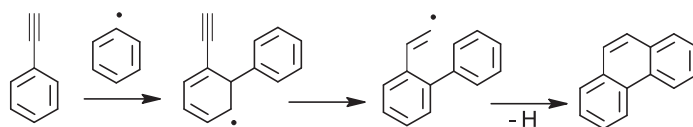
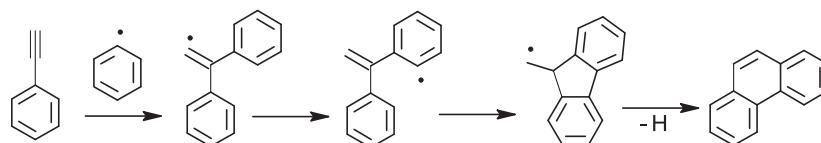
Table 3

Selected elementary reaction rate constants in the modified Arrhenius equation form of $A T^n \exp(-E/RT)$ with the units of cal, K, mol, cm and s. “p.w.” represents the present work rate constants estimated using DFT and TST. “p.w., [N]” indicates that the reaction was taken from Ref. N, and the rate constant was evaluated in this work. R5 represents five membered ring on PAHs, and “.” radical site. P2 and P2' are biphenyl and its radical. Only those H-abstraction reactions, whose rate constants have been evaluated in this work, are listed with species nomenclature adopted from [56].

No.	Reaction	A	n	E	Ref.
Naphthalene (A2) formation					
N1	$A1C_2H^{\cdot} + C_2H_2 \rightarrow A1C_2HC_2H_2$	2.30×10^{04}	2.530	2860	p.w.
-N1	$A1C_2HC_2H_2 \rightarrow A1C_2H^{\cdot} + C_2H_2$	4.80×10^{13}	0.456	43,186	p.w.
N2	$A1C_2HC_2H_2 \rightarrow A1C_2HC_2H_2u$	2.70×10^{12}	0.040	5092	p.w.
-N2	$A1C_2HC_2H_2u \rightarrow A1C_2HC_2H_2$	3.89×10^{12}	0.029	4307	p.w.
N3	$A1C_2H_2C_2Hu \rightarrow A2^{\cdot}$	1.20×10^{12}	-0.003	4484	p.w.
-N3	$A2^{\cdot} \rightarrow A1C_2HC_2H_2u$	2.98×10^{12}	0.681	59,594	p.w.
N4	$A1C_2H^{\cdot} + C_2H_2 \rightleftharpoons A1C_2HC_2H + H$	1.80×10^{19}	-1.670	18,800	[2]
N5	$A1C_2HC_2H + H \rightarrow A1C_2HC_2H_2u$	1.20×10^{09}	1.530	3695.5	p.w.
-N5	$A1C_2HC_2H_2u \rightarrow A1C_2HC_2H + H$	1.97×10^{11}	0.986	39,642	p.w.
N6	$A2^{\cdot} + C_2H_2 \rightleftharpoons A2C_2H_2$	1.58×10^{04}	2.550	3951.6	p.w., [2]
N7	$A2C_2H_2 \rightleftharpoons A2R5 + H$	3.48×10^{11}	0.181	15,551	p.w., [2]
N8	$A2 + CH_3 \rightarrow A2^{\cdot} + CH_4$	3.99×10^{-01}	3.933	11,771	p.w.
-N8	$A2^{\cdot} + CH_4 \rightarrow A2 + CH_3$	4.48×10^{-02}	4.248	4277	p.w.
N9	$A2 + C_3H_3 \rightarrow A2^{\cdot} + C_3H_4$	9.538×10^0	3.529	24449.2	p.w.
-N9	$A2^{\cdot} + C_3H_4 \rightarrow A2 + C_3H_3$	1.36×10^0	3.761	1088.91	p.w.
N10	$A2 + H \rightarrow A2^{\cdot} + H_2$	2.45×10^{08}	1.884	9829.5	p.w., [2]
-N10	$A2^{\cdot} + H_2 \rightarrow A2 + H$	4.90×10^{04}	2.467	2926.4	p.w., [2]
N11	$A1^{\cdot} + C_4H_4 \rightarrow A2 + H$	1.26×10^{04}	2.610	1649.2	[54]
N12	$C_5H_5 + C_5H_5 \rightarrow A2 + 2H$	6.39×10^{29}	-4.030	35205.5	[1]
N13	$C_6H_5CH_2 + C_3H_3 \rightleftharpoons C_{10}H_{10}$	1.51×10^{75}	-17.90	39,600	[45]
N14	$C_6H_5CH_3 + C_3H_2 \rightleftharpoons C_{10}H_{10}$	1.51×10^{75}	-17.90	39,600	[45]
N15	$C_{10}H_{10} + H \rightleftharpoons C_{10}H_9 + H_2$	1.00×10^{13}	0	0	[45]
N16	$C_{10}H_9 + H \rightleftharpoons C_{10}H_{10}$	1.00×10^{13}	0	0	[45]
N17	$C_{10}H_9 + H \rightleftharpoons A2 + H_2$	1.00×10^{13}	0	0	[45]
N18	$A2CH_3 + H \rightleftharpoons A2 + CH_3$	3.85×10^{06}	2.2	4163.5	[1]
Phenanthrene (A3) formation					
N19	$A2^{\cdot} + C_4H_4 \rightarrow A3 + H$	1.26×10^{04}	2.610	1649.2	[54]
N20	$A1C_2H^{\cdot} + A1^{\cdot} \rightarrow A3 + H$	8.60×10^0	3.066	5148	p.w., [2]
N21	$A1C_2H + A1^{\cdot} \rightarrow A3 + H$	1.40×10^{01}	3.052	5451	p.w., [2]
N22	$A1C_2H + A1^{\cdot} \rightarrow A3 + H$	4.82×10^{01}	3.080	7302.2	p.w.
N23	$A2CH_2 + C_3H_3 \rightarrow A3 + 2H$	1.51×10^{75}	-17.8	39600	[52]
N24	$A2CH_2 + C_3H_5 \rightarrow A3 + 2H_2$	2.50×10^{12}	0	0	[52]
N25	$C_6H_5CH_2 + C_6H_5CH_2 \rightarrow C_{14}H_{14} + 2H$	5.01×10^{12}	0	454	[45]
-N25	$C_{14}H_{14} \rightarrow C_6H_5CH_2 + C_6H_5CH_2$	1.40×10^{13}	0	59,751	[45]
N26	$C_{14}H_{14} \rightleftharpoons C_{14}H_{12} + H_2$	1.50×10^{15}	0	84,700	[45]
N27	$C_{14}H_{14} + H \rightarrow C_{14}H_{12} + H + H_2$	2.50×10^{15}	0	84,700	[45]
N28	$C_{14}H_{12} \rightarrow A3 + H_2$	8.00×10^{13}	0	8235	[45]
N29	$C_6H_7 + C_5H_5 \rightarrow A3 + 2H$	6.39×10^{29}	-4.030	35205.5	[1]
N30	$Biphenyl^{\cdot} + C_2H_2 \rightarrow A3 + H$	2.6×10^{04}	2.55	5644.4	p.w., [1]
N31	$A2R5 + C_2H_2 \rightleftharpoons A3^{\cdot}$	3.29×10^{06}	2	3161.1	[1]
N31	$A3 + CH_3 \rightarrow A3^{\cdot} + CH_4$	1.17×10^{-02}	4.177	12,246	p.w.
-N32	$A3^{\cdot} + CH_4 \rightarrow A3 + CH_3$	3.13×10^{-03}	4.486	6563.7	p.w.
N33	$A3 + C_3H_3 \rightarrow A3^{\cdot} + C_3H_4$	3.27×10^{-01}	3.768	24,434	p.w.
-N33	$A3^{\cdot} + C_3H_4 \rightarrow A3 + C_3H_3$	1.12×10^{-01}	3.994	2904.7	p.w.
N34	$A3 + H \rightarrow A3^{\cdot} + H_2$	6.12×10^{07}	1.921	9824.5	p.w., [2]
-N34	$A3^{\cdot} + H_2 \rightarrow A3 + H$	2.96×10^{04}	2.498	4751.6	p.w., [2]
Pyrene (A4) formation					
N35	$A2C_2H^{\cdot} + C_4H_4 \rightarrow A4 + H$	3.33×10^{33}	-5.700	12,750	[52]
N36	$C_6H_5CH_2 + C_9H_7 \rightarrow A4 + 2H_2$	2.00×10^{12}	0	2000	[52]
N37	$A2CH_2CH_2 + C_5H_5 \rightarrow A4 + 2H_2$	5.00×10^{13}	0	2000	[52]
N38	$C_9H_7 + C_9H_7 \rightarrow A4 + C_2H_2 + H_2$	6.39×10^{29}	-4.030	35205.5	[52]
N39	$A3^{\cdot} + C_2H_2 \rightarrow A4 + H$	1.28×10^{06}	2.050	1931.17	[1]
N40	$A4 + CH_3 \rightarrow A4^{\cdot} + CH_4$	7.98×10^{-01}	3.933	11,771	p.w.
-N40	$A4^{\cdot} + CH_4 \rightarrow A4 + CH_3$	4.48×10^{-02}	4.248	4277	p.w.
N41	$A4 + C_3H_3 \rightarrow A4^{\cdot} + C_3H_4$	$1.91 \times 10^{+01}$	3.529	24449.2	p.w.
-N41	$A4^{\cdot} + C_3H_4 \rightarrow A4 + C_3H_3$	1.36×10^0	3.761	1088.91	p.w.
N42	$A4 + H \rightarrow A4^{\cdot} + H_2$	4.90×10^{08}	1.884	9829.5	p.w., [2]
-N42	$A4^{\cdot} + H_2 \rightarrow A4 + H$	4.90×10^{04}	2.467	2926.4	p.w., [2]
Benzo[e]pyrene (A5), benzo[ghi]perylene (A6) and coronene (A7) formation					
N43	$A4^{\cdot} + C_4H_4 \rightarrow A5 + H$	1.26×10^{04}	2.610	1649.2	[54]
N44	$A4C_2H^{\cdot} + C_2H_2 \rightleftharpoons A5^{\cdot}$	1.87×10^{07}	1.787	3262	[53]
N45	$Perylene^{\cdot} + C_2H_2 \rightarrow A6 + H$	1.87×10^{07}	1.787	3262	[53]
N46	$A5^{\cdot} + C_2H_2 \rightarrow A6 + H$	1.87×10^{07}	1.787	3262	[53]
N47	$A6^{\cdot} + C_2H_2 \rightarrow A7 + H$	1.87×10^{07}	1.787	3262	[53]
Other hydrogen abstraction reactions by H, CH₃ and C₃H₃					
N48	$P2 + CH_3 \rightarrow P2^{\cdot} + CH_4$	1.49×10^{-01}	4.002	11934.1	p.w.
-N48	$P2^{\cdot} + CH_4 \rightarrow P2 + CH_3$	1.27×10^{-02}	4.309	5108.7	p.w.
N49	$P2 + C_3H_3 \rightarrow P2^{\cdot} + C_3H_4$	3.32×10^0	3.571	24,292	p.w.

Table 3 (continued)

No.	Reaction	A	n	E	Ref.
-N49	$P2^{\cdot} + C_3H_4 \rightarrow P2 + C_3H_3$	3.60×10^{-01}	3.795	1619.8	p.w.
N50	$P2 + H \rightarrow P2^{\cdot} + H_2$	1.61×10^{08}	1.912	9897.6	p.w., [2]
-N50	$P2^{\cdot} + H_2 \rightarrow P2 + H$	2.46×10^{04}	2.486	3682.2	p.w., [2]
N51	$C_9H_8 + CH_3 \rightarrow C_9H_7 + CH_4$	2.75×10^0	3.614	3380.1	p.w.
-N51	$C_9H_7 + CH_4 \rightarrow C_9H_8 + CH_3$	1.10×10^{-02}	3.883	29,788	p.w.
N52	$C_9H_8 + C_3H_3 \rightarrow C_9H_7 + C_3H_4$	5.15×10^{-03}	4.376	8021.8	p.w.
-N52	$C_9H_7 + C_3H_4 \rightarrow C_9H_8 + C_3H_3$	2.63×10^{-03}	4.563	18,582	p.w.
N53	$C_9H_8 + H \rightarrow C_9H_7 + H_2$	7.61×10^{07}	1.841	-126.5	p.w., [2]
-N53	$C_9H_7 + H_2 \rightarrow C_9H_8 + H$	5.46×10^{04}	2.378	26,891	p.w., [2]
N54	$C_5H_6 + CH_3 \rightarrow C_5H_5 + CH_4$	3.07×10^0	3.579	3565.6	p.w.
-N54	$C_5H_5 + CH_4 \rightarrow C_5H_6 + CH_3$	1.78×10^{-01}	3.708	26869.2	p.w.
N55	$C_5H_6 + C_3H_3 \rightarrow C_5H_5 + C_3H_4$	4.89×10^{-03}	4.326	8587.6	p.w.
-N55	$C_5H_5 + C_3H_4 \rightarrow C_5H_6 + C_3H_3$	1.82×10^{-04}	4.372	16043.9	p.w.
N56	$C_5H_6 + H \rightarrow C_5H_5 + H_2$	5.995×10^{07}	1.876	70.5	p.w., [2]
-N56	$C_5H_5 + H_2 \rightarrow C_5H_6 + H$	6.27×10^{03}	2.273	23,984	p.w., [2]
N57	$A1C_2H + CH_3 \rightarrow A1C_2H^{\cdot} + CH_4$	8.81×10^{-01}	3.843	12253.8	p.w.
-N57	$A1C_2H^{\cdot} + CH_4 \rightarrow A1C_2H + CH_3$	1.95×10^{-01}	4.154	3700.9	p.w.
N58	$A1C_2H + C_3H_3 \rightarrow A1C_2H^{\cdot} + C_3H_4$	1.21×10^{01}	3.443	24836.1	p.w.
-N58	$A1C_2H^{\cdot} + C_3H_4 \rightarrow A1C_2H + C_3H_3$	3.43×10^0	3.671	436.1	p.w.
N59	$A1C_2H + H \rightarrow A1C_2H^{\cdot} + H_2$	1.35×10^{08}	1.890	10,767	p.w., [2]
-N59	$A1C_2H^{\cdot} + H_2 \rightarrow A1C_2H + H$	5.38×10^{04}	2.469	2827.1	p.w., [2]
N60	$A1 + CH_3 \rightarrow A1^{\cdot} + CH_4$	4.42×10^0	3.916	11,463	p.w.
-N60	$A1^{\cdot} + CH_4 \rightarrow A1 + CH_3$	3.94×10^{-01}	4.212	4145	p.w.
N61	$A1 + C_3H_3 \rightarrow A1^{\cdot} + C_3H_4$	2.27×10^{01}	3.517	24,065	p.w.
-N61	$A1^{\cdot} + C_3H_4 \rightarrow A1 + C_3H_3$	2.58×10^0	3.730	899.32	p.w.

Pathway P_{A3}1Pathway P_{A3}2Fig. 2. Two pathways for the formation of A3 from A1C₂H and A1[•]. P_{A3}1 is the pathway for reaction N21 and P_{A3}2 is the pathway for reaction N22.

can be approximated with the rate constant for that reaction on a smaller PAH molecule, as they are not significantly different. The thermodynamic data for the newly added species were taken from [1,53].

3.2. Validation for premixed laminar flames

This section provides the comparison between the simulation results obtained using the new mechanism and the experimental data for the premixed laminar flames listed in Table 1. For some flames, the simulations were also carried out with Mechanisms 1 and 2 to show the improvements obtained with the new mechanism.

3.2.1. N-heptane and iso-octane flames

Figures 3 and 4 show the profiles of the mole fractions of small chemical species (smaller than PAHs) as a function of the height above the burner (HAB) formed in the premixed laminar flames of n-heptane and iso-octane (PF1a and PF1b in Table 1), respectively. The simulations were conducted with the new mechanism and the base mechanism without any soot model. In PF1a, the computed profiles of the small chemical species with the new

mechanism differed only moderately from those obtained using the base mechanism. The agreements between the computed and the observed species profiles are reasonable. For A1, the agreement with the new mechanism is improved as compared to the base mechanism at higher HABs, while the maximum values are under-predicted by around 30%. In PF1b for the iso-octane flame (Fig. 4), the mole fraction of C₃H₄ is under-predicted for both the new and the base mechanism, which may be responsible for the under-predicted A1 mol fractions at lower HABs, and thus requires further investigation.

The comparison between the computed and the observed ignition delay times for several fuels such as n-heptane, iso-octane, toluene and benzene are shown in the Supplementary material of this paper, where good agreements between them are obtained.

Two n-heptane premixed laminar flames (PF2a and PF2b), where the PAH profiles up to pyrene are available, were simulated using the new mechanism to predict the profiles of large PAHs. Figures 5 and 6 show the profiles of A1–A4 together with A5–A7. These flames were also simulated using Mechanism 1 [1] and Mechanism 2 [2] to show the improvements achieved with the new mechanism. In the experimental paper [7], it is mentioned that the experimental mole fraction data within a few millimeters

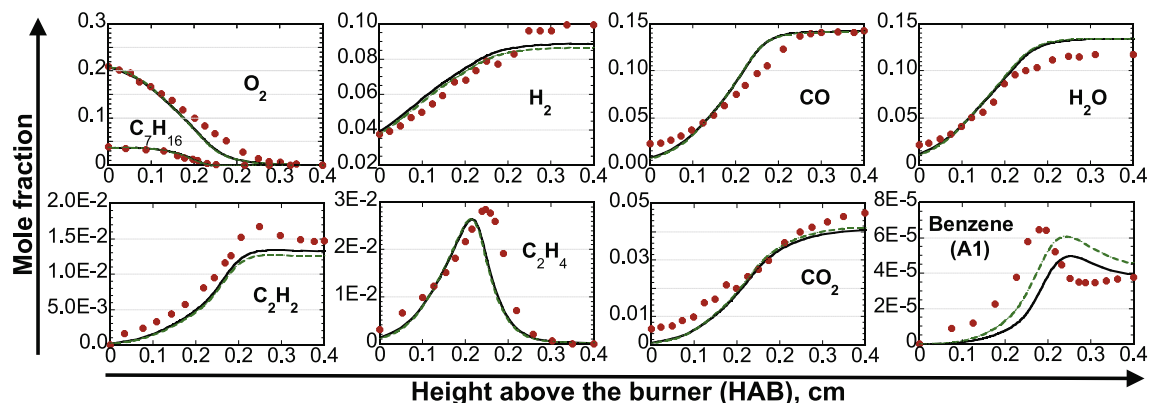


Fig. 3. Mole fraction profiles in C_7H_{16} premixed flame (PF1a) as a function of height above the burner. The simulation results from the new and the base mechanism are shown by solid and dotted lines, respectively.

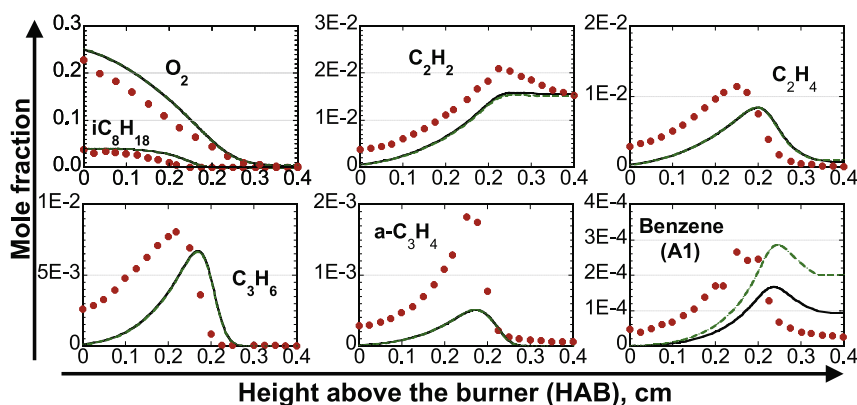


Fig. 4. Mole fraction profiles in iC_8H_{18} premixed flame (PF1b) as a function of height above the burner. The simulation results from the new and the base mechanism are shown by solid and dotted lines, respectively.

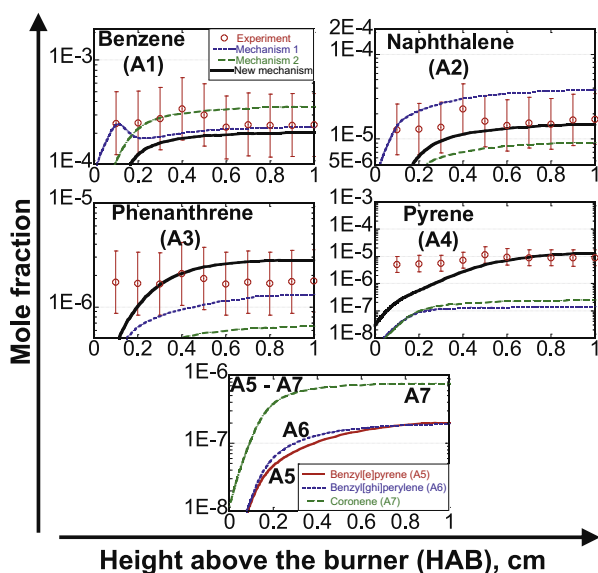


Fig. 5. Comparison of PAH profiles for n-heptane flame (PF2a) from simulation and experiment [7].

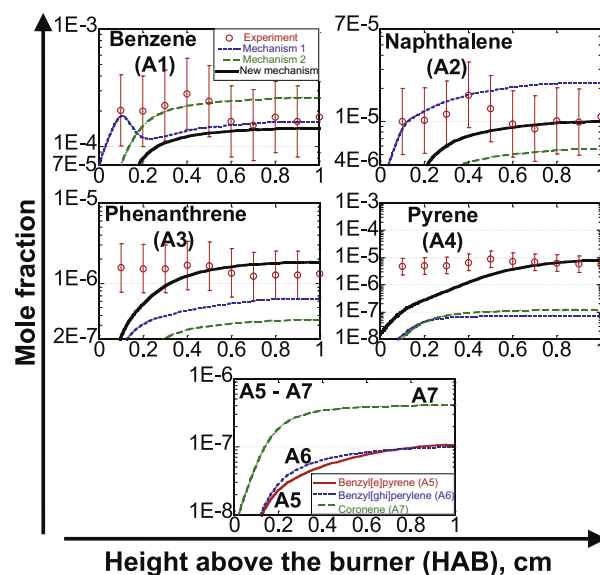


Fig. 6. Comparison of PAH profiles for n-heptane flame (PF2b) from simulation and experiment [7].

above the burner surface may be considered “questionable” due to possible sampling probe-burner surface interactions. Thus, only the comparison at the large HABs may be meaningful.

The mole fraction of A1 is predicted reasonably well with all the three mechanisms within 45% as compared to the experimental

data at large HABs. Significant improvements have been obtained with the present mechanism for larger PAHs, especially for A3 and A4, at large HABs. With Mechanisms 1 and 2, at higher HABs, the concentration of A4 (which is an important PAH for soot

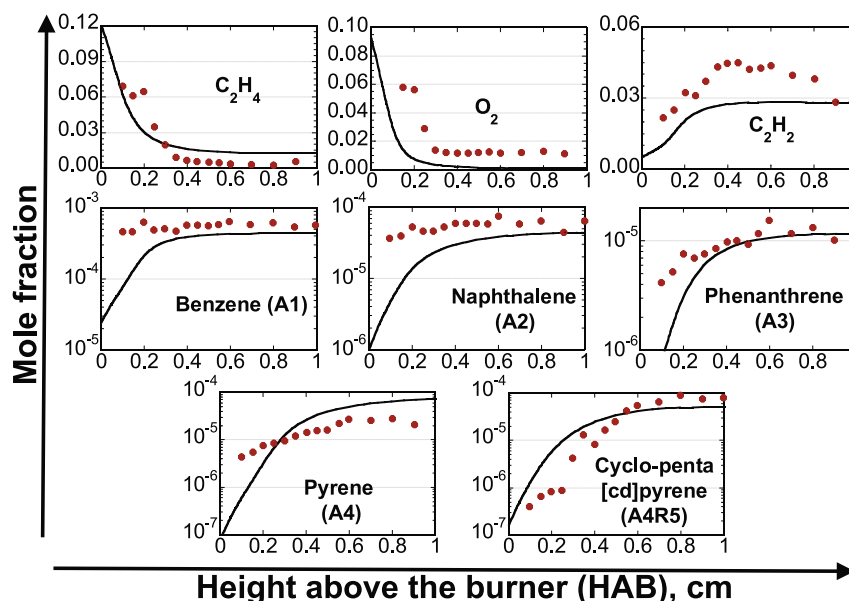


Fig. 7. Mole fraction profiles in the ethylene flame (PF3) as a function of height above the burner.

inception) is under-predicted by a factor of about 100. In both flames, with the new mechanism, the concentration of A3 is slightly over-predicted, while that of A4 is very well-predicted beyond HAB = 0.5 cm.

Only the computed mole fraction profiles of A5–A7 are presented in Figs. 5 and 6 due to the absence of experimental data on these PAHs. The mole fraction of A7 is higher than those of A5 and A6 due to the absence of its consumption pathways in the mechanism, since A7 is the largest chemical species present in the mechanism.

The simulation results presented in the rest of the paper will only be based on the new mechanism, as the base mechanism is shown to significantly under-predict the concentrations of large PAHs in this section.

3.2.2. Ethylene and benzene flames

Figures 7 and 8 provide the profiles of small chemical species together with PAHs in ethylene (PF3) and benzene (PF4) flames, respectively, where the experimental data for large PAHs are available. The concentrations of the PAHs for these flames are predicted satisfactorily. It has been pointed out in [7,57,58] that an agreement between the computed and the observed PAH profiles within an order of magnitude can be considered satisfactory, given that a significant amount of uncertainty is involved in the measurement of the concentrations of the large PAHs. In the ethylene flame, the mole fraction profiles of the PAHs (A1–A4 and A4R5) are well-predicted. At higher HABs in the benzene flame, the mole fractions of the PAHs larger than A3 are slightly over-predicted.

Note that, in Table 3, the reactions involving the formation of 6-membered aromatic rings (N11, N19, N21, N28, N30, N35, N39, N43 and N45–47) were assumed to be irreversible, as the breakage of an aromatic ring requires high activation energy [29]. This assumption was tested in the benzene (PF4) flame. The maximum values of the PAH concentrations were found to change by 0.1–2.5% only, and therefore, the reverse reactions may be neglected.

3.3. Validation for counterflow diffusion flames

Having the modified mechanism which can predict reasonably well the PAH mole fractions in the premixed flames, the present mechanism has been tested for the counterflow diffusion flames,

especially focused on the cross-linking effect by fuel mixing. Under the experimental conditions of [3], the iso-octane and n-heptane flames are non-sooting, whereas toluene has a high sooting tendency. Therefore, it can be expected that, if the toluene ratio is increased in iso-octane/toluene or n-heptane/toluene mixture, the concentration of PAHs in a flame would increase. It was observed experimentally [3] that with the increasing toluene ratio in these fuel mixtures, PAH concentration firstly increases, and then decreases with further increase in the toluene ratio. In other words, PAHs in iso-octane – toluene (and n-heptane – toluene) flame exhibit a synergistic effect with the variation in the toluene ratio in the fuel mixture. This effect arises due to crosslinking between gasoline surrogate fuel constituents. For these fuel mixtures, such a synergistic effect was not exhibited in the LII signals (i.e. soot particles). While, in literatures, this effect has also been reported for soot particles formed during the combustion of mixture fuels in shock tubes [59–61] and in flames [20,36,62]. This may be attributed to the fact that in the counterflow experiments [3], n-heptane and iso-octane were non-sooting for the specified flow conditions.

Figure 9 shows experimental and computed normalized maximum mole fractions of PAHs in SF and SFO flames of iso-octane – toluene and n-heptane – toluene mixtures. For each flame, the experimental data points were obtained by taking the maximum value of laser-induced fluorescence (LIF) signal in the flame, and then normalizing it with that observed for the pure toluene flame (toluene ratio, $R_T = 1$) [3]. Experimental data for individual PAHs were not available because of the broad excitation and emission characteristics of PAHs, while the reported data measured at 400 nm (excitation at 283 nm) may represent PAHs with 1–4 rings. According to the findings of Ossler et al. [63], the spectra of individual PAHs were observed at different temperatures and concentrations. It was shown that A2 and A4 can show maximum fluorescence intensity near 400 nm at high temperature and concentration. Note that the observed LIF signals in a flame can have contributions from benzene and several PAHs, and therefore, direct comparison of the observed and the computed normalized mole fractions of the individual PAHs are limited.

With the new mechanism, it can be seen in Fig. 9 that all the major PAH molecules (A1, A2, A3 and A4) exhibit synergistic effect in both SF as well as SFO flames of iso-octane – toluene and

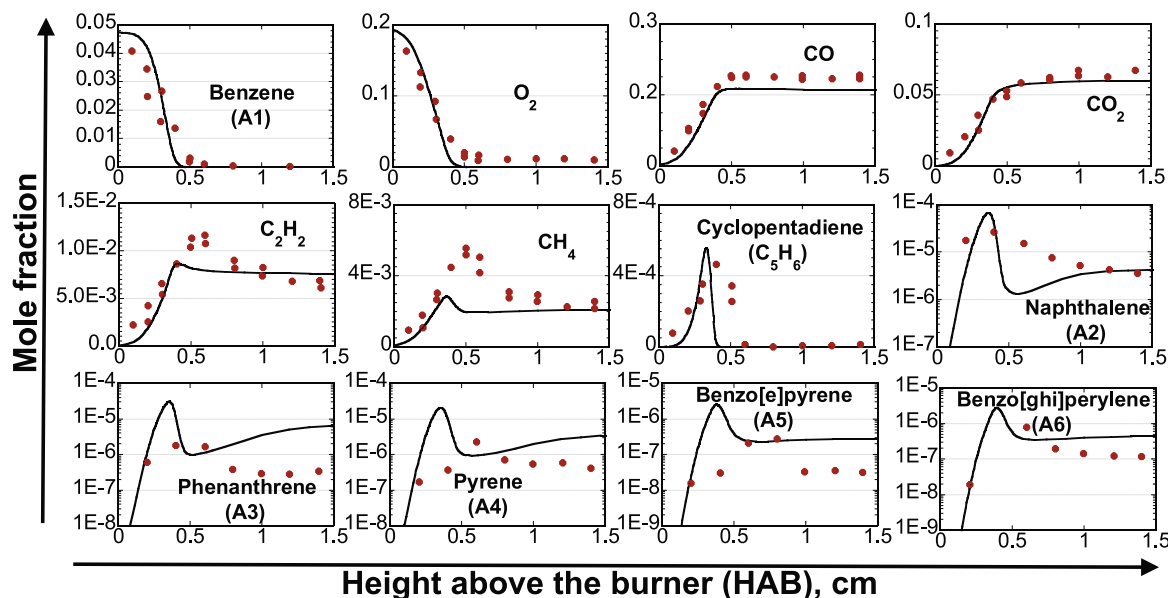


Fig. 8. Mole fraction profiles in the benzene flame (PF4) as a function of height above the burner.

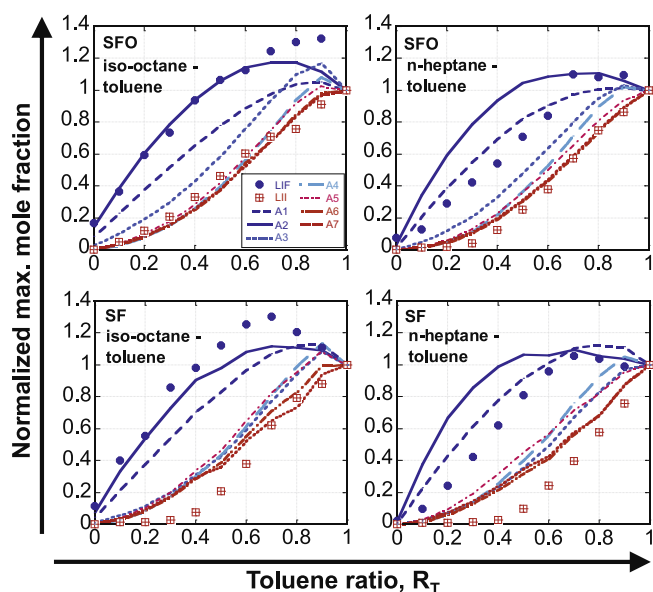


Fig. 9. Prediction of the synergistic effect shown by PAHs in the counterflow flames of n-heptane/toluene and iso-octane/toluene mixtures in SF as well as SFO conditions. The experimental data points represent the maximum values of normalized laser induced fluorescence (LIF) signals and laser induced incandescence (LII) signals. A1–A7 are different PAH molecules, whose IUPAC names and structures can be obtained from Fig. 1.

n-heptane – toluene mixtures (except A3 in n-heptane/toluene SF flames). Out of A5, A6 and A7, only A5 showed synergistic effect in the flames of iso-octane/toluene mixtures, and none of them showed this effect in n-heptane/toluene flames. Further discussion on the profiles of these PAHs are presented later.

It has been shown in [3] for both types of flames that, with Mechanism 1 [1], such a synergistic effect for PAHs (except for A1 out of A1–A4) was not obtained. With Mechanism 2 [2], A2 did not show this effect for both SF and SFO flames. In the SF flames, this effect was over-predicted by a maximum factor of around 2 for A3 and around 6.5 for A4. In SFO flames, the synergistic effect was reasonably well predicted for A1, A3 and A4 in iso-octane – toluene mixture flames, but in n-heptane – toluene mixture flames, only A1 showed this effect. In the experiment [3], the maxima for the LIF signals of PAHs exhibited the synergistic effect for $R_T \in [0.7, 0.9]$ for both SF and SFO flames, which is same as the range shown by computed PAHs in Fig. 9, especially for A2.

In the two SFO cases, the profiles of A2 are able to predict well the value of $R_T (=0.9)$, at which the LIF signal intensities show their maxima. In the SF cases, the maxima in LIF signal intensities occur at $R_T = 0.7$. Interestingly, the profiles of A1 attain their maxima at the same R_T . As mentioned above, the reported LIF data were measured at 400 nm. The fluorescence spectra of A1 and A2 show their maximum intensities at around 280 nm and 330 nm, respectively. However, it has been shown in [63] that the location of maxima in the fluorescence spectra of a PAH depends on its concentration and temperature, and it shifts towards higher wavelengths with increasing concentration of aromatic hydrocarbons. Since A1 is present in higher concentrations than the PAHs, it can indeed be responsible for the observed negative slope of the LIF intensity profiles after $R_T = 0.7$ in the SF cases. An experimental investigation is required in the future where the LIF signals from a synthetic mixture of A1 with different PAHs at flame conditions can be observed in order to determine the contribution of A1 to the LIF signal intensities near 400 nm.

Note that in both types of flames, the profiles of large PAHs (A4–A7) are significantly different from A1 and A2. They show a very slow start until R_T equals to 0.2, which could be due to the resistance of iso-octane and n-heptane in forming larger PAHs, and subsequently soot, in high concentrations. Similar behavior has been shown by the laser-induced incandescence (LII) signals from these flames representing soot volume fraction. The signals were minimal for $R_T \leq 0.4$ for SF flames. This indicates that there exists a tolerance in terms of toluene mixing to either n-heptane or iso-octane, below which the effect of toluene on soot formation can be regarded as small [3].

The similarity between the profiles of large PAHs (A4–A7) and LII signals implies the involvement of such PAHs in the formation/inception of soot particles. In literatures, soot particles are generally modeled to inception through the dimerization of PAHs ranging from A1 to A5 [41,58,64–66]. In some research works involving the modeling of turbulent or non-premixed flames, C_2H_2 has been assumed to inception soot [67,68]. Such models may

be suitable for practical applications to predict soot observables. However, in order to improve our understanding on soot formation pathways, modeling efforts are required where the inception and growth of soot particles can take place through PAHs larger than A4. The simulated profiles of large PAHs (A4–A7), as shown in Fig. 9, exhibit negligible synergistic effect. This may get enhanced when soot formation module is included in the mechanism. The effect of soot formation on the synergistic behavior of large PAHs forms a part of the future study.

3.3.1. Kinetic pathways

In order to understand the synergistic behavior in the formation of PAHs for gasoline surrogate fuels, the rate-of-production analysis [37] was conducted to determine the reactions responsible for the formation of A1–A7 in the SF flame of iso-octane/toluene mixture at $R_T = 0.9$. This analysis was carried out at different temperatures (or locations) in the flame as the major pathways for PAH formation may differ with flame location. Figure 10 presents the pathways, important for the formation of large PAHs starting from benzene. The formation pathways for benzene are not shown in this figure, though it was found that it mainly formed through propargyl (C_3H_3) recombination in the case of aliphatic flames and through the reaction $C_6H_5CH_3 + H \rightleftharpoons C_6H_6 + CH_3$ in the case of toluene-rich flames.

With the new rates for the formation of A2 from A1 (reactions N1–N5), the concentration of A2 increased in the case of aliphatic flames, as the new rates for these reactions were higher than their rates present in the literature [56]. Furthermore, the new rates for the formation of acenaphthylene, A2R5 (reactions 6 and 7) were found to be less than the rates present in the literature [56] by a factor of 10, which led to decrease in the conversion of A2 to A2R5. The rate constants for several important PAH reactions on which DFT calculations were performed along with those reactions, which contribute significantly to the formation of PAHs during the combustion of large hydrocarbons including aromatic compounds are provided in Table 3.

For the formation of A2, A3 and A4, two radical species, benzyl ($C_6H_5CH_2$) and indenyl (C_9H_7), were found to be actively involved. Note that both the species are resonantly stabilized radicals (RSRs). The benzyl radicals are formed from toluene dehydrogenation and the indenyl radicals are formed either from the reaction between C_2H_2 and C_6H_5CH (a bivalent toluene radical) or through the addition of C_3 species on A1 (or A1 $\cdot$). Along with these, other RSRs such as C_3H_3 , C_3H_5 (allyl) and C_5H_5 contribute significantly to the PAH growth, as can be seen in Table 3. These radicals can exist in a flame in substantial concentration due to their high stability, which arises from resonance and electron delocalization [69]. For example, the high concentrations of RSRs, especially C_3H_3 and C_5H_5 , in the combustion environments of several fuels such as propene [70], acetylene [71], cyclopentene [72], 1,3-pentadiene [73], toluene [74], n-heptane [75] and C_6H_{12} [76] have already been reported in the literatures using theoretical and experimental approaches. RSRs can act as a sink for H atoms [74], and thus can reduce the radical pool in flames. Once RSRs are formed, these long-lived intermediates can combine together (for example, C_3H_3 recombination to form A1 and C_5H_5 recombination to form A2) or react with other chemical species (for example, reaction between C_5H_5 and CH_3 to form A1, and addition of C_3H_3 or C_3H_5 on A2CH $_2$ to form A3) to increase benzene and PAH concentrations in flames.

One of the gasoline surrogate fuels, toluene, is a well-known RSR precursor [74]. The concentrations of RSRs, such as benzyl and phenoxy (C_6H_5O), are influenced with a high sensitivity when another fuel such as n-heptane is mixed with toluene. Previous modeling studies including those on n-heptane/toluene mixtures [74,77,78] have already highlighted the prominent role of RSRs in flames.

In this work, it will be shown later that RSRs such as C_3H_3 , C_9H_7 and $C_6H_5CH_2$ also play significant roles in causing synergistic effect of PAHs in the flames of mixture fuels. In a previous study on the flames of ethylene-propane mixtures [20], the observed synergistic effect of PAHs was explained based on the variation in C_3H_3 concentration. In a recent review on soot particles [69], the role

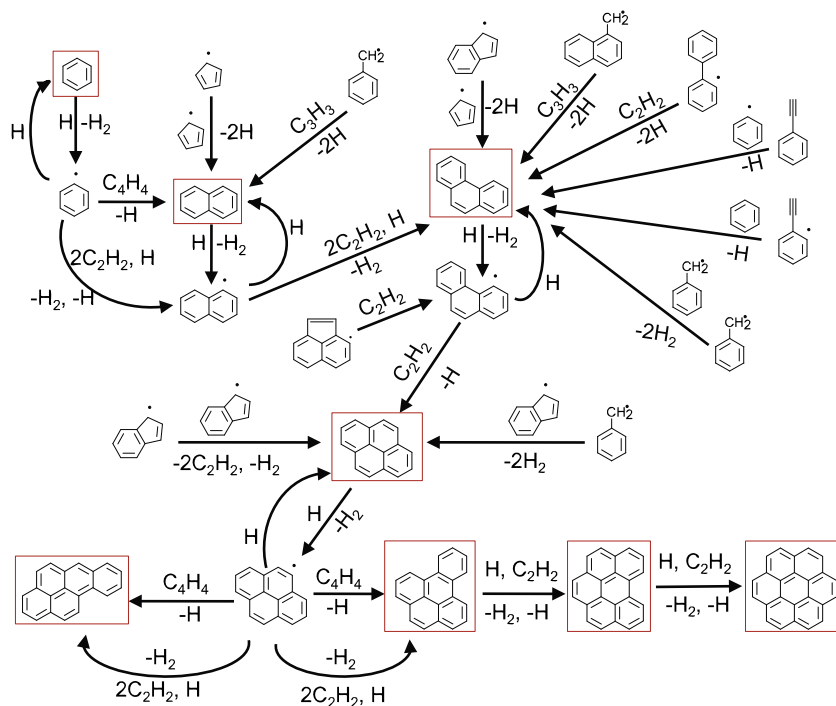


Fig. 10. Major reaction pathways for the formation of large PAHs. For simplicity, the isomers of some of the radical species are not shown.

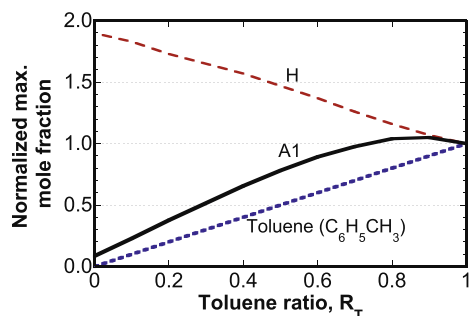


Fig. 11. Normalized maximum mole fractions of the species involved in the pathway responsible for the synergistic effect of A1 in the SFO flame of iso-octane/toluene mixture for different R_T values.

of stabilized aromatic radicals in the inception of soot particles has been highlighted. Therefore, such species not only promote PAH formation and growth, but can also lead to soot formation. Clearly, an accurate representation of the RSR chemistry in a chemical mechanism is the key for understanding complex phenomena such as synergistic effect and soot formation in the flames of gasoline surrogate fuel mixtures.

3.3.2. Synergistic effect

This section provides the results of the mechanism analysis conducted on the new mechanism using the test procedure described in Section 2.3 to determine the reactions and the gas-phase species responsible for the synergistic behavior of four PAHs, A1–A4, in the soot formation/oxidation (SFO) flames of iso-octane/toluene mixtures.

3.3.2.1. Benzene (A1). The reactions for the formation of A1 in the mechanism involve several species such as C_2H_2 , C_3H_3 , C_3H_4 , C_3H_5 , C_4H_3 , H, $A1CH_3$ and linear and cyclic C_6 molecules. Out of 16 reactions forming benzene and phenyl, only one of them leads to its synergistic behavior: $C_6H_5CH_3(\text{toluene}) + H \rightleftharpoons A1 + CH_3$. Figure 11 shows the normalized maximum mole fractions of the species involved in this reaction by varying R_T . With increasing R_T , the concentration of toluene increases linearly, while that of H atoms decreases nearly linearly. The opposite near linear trends in H and toluene could lead to the parabolic variation (and synergistic effect) in A1, since the reaction rate for the formation of A1 is proportional to the product of the concentrations of H and toluene. Note that this synergistic behavior of A1 has been explained previously in [3].

3.3.2.2. Naphthalene (A2). The reactions forming A2 mainly involve: (a) the addition of C_2H_2 , C_2H_4 , C_3H_3 , C_4H_3 and C_4H_4 on smaller PAHs, (b) the addition of H atom on $A2^{\cdot}$ ($A2$ radical) by H, H_2 , CH_4 , C_3H_4 and H_2O , (c) the replacement of methyl group of methyl-naphthalene by H atom ($A2CH_3 + H \rightleftharpoons A2 + CH_3$), and (d) the self-addition of cyclopentadienyl radicals (C_5H_5). Some of these reactions are already listed in Table 3. Using the test procedure described above, those reactions responsible for the synergistic behavior of A2 were determined. Table 4 provides the list of such reactions (Pathways P1–P5).

The normalized maximum mole fractions of the species involved in these reactions by varying R_T are provided in Fig. 12. It can be seen that the synergistic effect in A2 comes from those reactions in which (a) one of the reactants shows the synergistic effect (P1, $A2CH_3$), or (b) the reactants show opposite trends (P2–P5). In Table 4, Pathway P1, which involves the replacement of CH_3 group in methyl-naphthalene ($A2CH_3$) by H atom is very similar to the reaction for $C_6H_5CH_3 + H \rightleftharpoons A1 + CH_3$ causing the synergistic effect in A1.

Pathway P2 involves the formation of A2 from its radical, $A2^{\cdot}$ (1-Naphthyl). Interestingly, the interaction of $A2^{\cdot}$ with species such as H_2 , CH_4 , and C_3H_4 and H_2O leads to the synergistic effect in A2. The major pathway for the formation of $A2^{\cdot}$ is presented in Table 4. It was shown in [3] that, with Mechanism 2, the synergistic effect of A2 was not obtained even though Pathway P2 was present in the mechanism. By updating the rate constants of the reactions involved in the formation of $A2^{\cdot}$ with those evaluated in this work, the synergistic effect for A2 could be obtained with Mechanism 2 as well. Therefore, reasonable estimates of the reaction rates in the mechanism are required to predict the synergistic behavior of a PAH molecule.

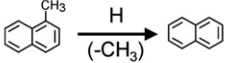
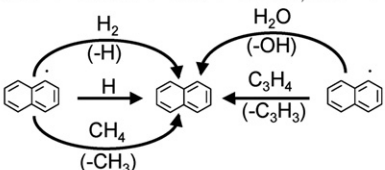
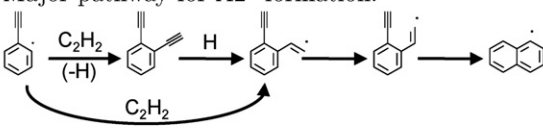
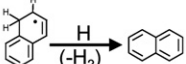
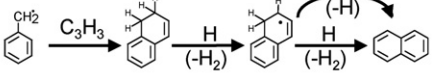
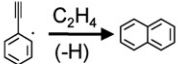
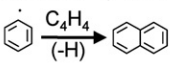
Pathway P3 involves $C_{10}H_9$ and H atoms as reactants. This reaction is an intermediate elementary step in the formation of A2 from $C_6H_5CH_2$ and C_3H_3 , and the pathway for the reaction between them is provided in Table 4. Pathway P4 involves the addition of C_4H_4 on a phenyl radical ($A1^{\cdot}$), while Pathway P5 involves the addition of C_2H_4 on a phenylacetylene radical ($A1C_2H^{\cdot}$).

3.3.2.3. Phenanthrene (A3). Table 5 lists the pathways P1–P5 leading to the synergistic effect in A3 along with the reactions for the formation of A3. The normalized maximum mole fraction profiles of the reactants involved in P1–P5 are shown in Fig. 13. In the mechanism, several reactions involve the addition of C_3 molecules such as C_3H_2 , C_3H_3 and C_3H_5 on $A2CH_2$. However, the opposite trends in the variation of normalized maximum mole fraction with R_T are shown only by $A2CH_2$ and C_3H_5 , as shown in Fig. 13, and the reaction between them (P1, Table 5) leads to the synergistic behavior of A3. The second pathway, P2, responsible for this effect involves the formation of A3 from $C_{14}H_{12}$ (which is formed through the self-addition of $C_6H_5CH_2$). The molecule $C_{14}H_{12}$ shows the synergistic effect, which is inherited from $C_6H_5CH_2$, which also exhibits this effect (see Fig. 14). Similar to A1 and A2, the synergistic behavior can also arise in A3 through the replacement of methyl group in methyl-phenanthrene ($A3CH_3$) by an H atom (P3, Table 5). Along with A2, methyl-naphthalene ($A2CH_3$) can also lead to the synergistic effect in A3 though the addition of C_3H_2 to it (P4 in Table 5). Pathway P5 involves the addition of C_2H_2 on a biphenyl radical ($P2^{\cdot}$). The variation of C_2H_2 mole fraction with R_T is near-linear, while that of $P2^{\cdot}$ is highly non-linear. $P2^{\cdot}$ is formed from the self-addition of $A1^{\cdot}$, and through the reaction between A1 and $A1^{\cdot}$. It can be seen that, except Pathways P1 and P5, the other three pathways involve synergistic effects in the reactants for A3 production.

3.3.2.4. Pyrene (A4). Out of several reactions forming A4, only three of them were found to be responsible for its synergistic effect and are listed in Table 6 as P1–P3. Pathway P1 has two reactants, $C_6H_5CH_2$ and C_9H_7 , where $C_6H_5CH_2$ shows the synergistic behavior (see Fig. 14) while the concentration of C_9H_7 increases monotonically with R_T . The synergistic effect in $C_6H_5CH_2$ comes from the reaction: $C_6H_5CH_3 + H \rightleftharpoons C_6H_5CH_2 + H_2$. This is very similar to the reaction: $C_6H_5CH_3 + H \rightleftharpoons A1 + CH_3$, which leads to the synergistic effect in A1. This indicates that, even though H atoms are not directly involved as a reactant, they can still influence the synergistic behavior of A4. The only reactant in P2, $A_3C_2H_2$, shows the synergistic effect (see Fig. 14), which directly propagates into A4. Note that this reaction is an intermediate elementary step involved in the formation of A4 from A3 through HACA mechanism. Pathway P3 involves the addition of C_2H_2 on $A3^{\cdot}$ (4-phenanthrenyl) followed by the ring closure and the loss of H atom in a single step. The concentration of C_2H_2 decreases nearly linearly with increasing R_T , while $A3^{\cdot}$ shows the synergistic behavior. Similar to P2, this pathway also forms a part of the HACA mechanism. Although HACA mechanism alone can predict the synergistic effect in A4,

Table 4

Reactions leading to the synergistic behavior of naphthalene (A2). Most of the reactions shown in the pathways are reversible and the arrows only show the direction of the progress of the reaction. The pathways indicate possible routes for the reactions to take place.

Pathway	Reactions
P1	$A2CH3 + H \rightleftharpoons A2 + CH3$ 
P2	$A2^{\bullet} + H \rightleftharpoons A2$, $A2^{\bullet} + H_2 \rightleftharpoons A2 + H$, $A2^{\bullet} + CH_4 \rightleftharpoons A2 + CH_3$ $A2^{\bullet} + C_3H_4 \rightleftharpoons A2 + C_3H_3$, $A2^{\bullet} + H_2O \rightleftharpoons A2 + OH$  Major pathway for A2 [•] formation: 
P3	$C10H9 + H \rightleftharpoons A2 + H_2$  Pathway for A2 formation through this route: 
P4	$A1^{\bullet} + C4H4 \rightarrow A2 + H$ 
P5	$A1C2H^{\bullet} + C2H4 \rightarrow A2 + H$ 

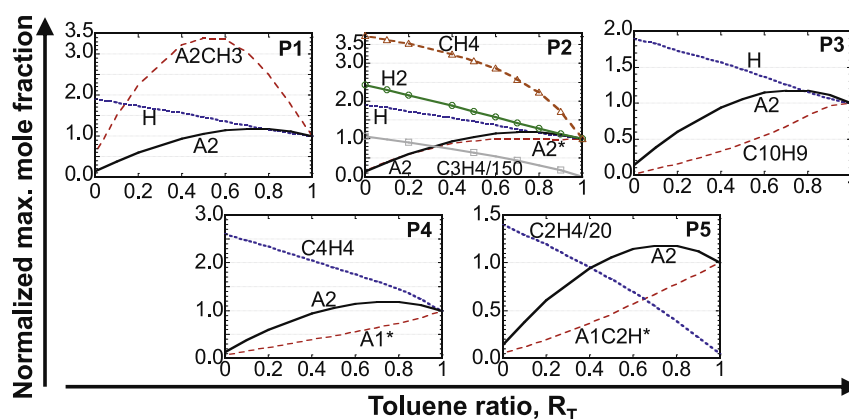


Fig. 12. Normalized maximum mole fractions of the species involved in the pathways responsible for the synergistic effect of A2 in the SFO flame of iso-octane/toluene mixture (Table 4 for P1–P5).

but, as mentioned previously, it may not fully account for the PAH growth in a reaction mechanism.

It is important to mention here that along with predicting the synergistic effect for PAHs, it is also necessary to predict well the ex-

tent of synergism as compared to experiment. The normalized LIF signals were found to increase by around 30% for SFO flames and around 10% for SF flames for $R_T \in [0.7, 0.9]$. This increase is reasonably well predicted with the new mechanism in both type of flames.

Table 5
Reactions leading to the synergistic behavior of phenanthrene (A3). Most of the reactions shown in the pathways are reversible and the arrows only show the direction of the progress of the reaction. The pathways indicate possible routes for the reactions to take place.

Pathway	Reactions
P1	$A2CH_2 + C_3H_5 \rightarrow A3 + 2H_2$ Pathway for A3 formation through this route:
P2	$C14H12 \rightleftharpoons A3 + H_2$ Pathway for A3 formation through this route:
P3	$A3CH_3 + H \rightleftharpoons A3 + CH_3$
P4	$A2CH_3 + C_3H_2 \rightarrow A3 + 2H$ Pathway for A3 formation through this route:
P5	$\text{biphenyl}^* + C_2H_2 \rightarrow A3 + H$

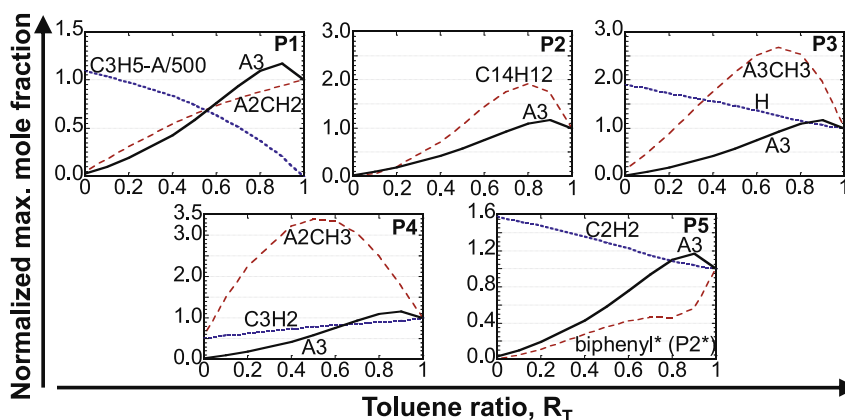


Fig. 13. Normalized maximum mole fractions of the species involved in the pathways responsible for the synergistic effect of A3 in the SFO flame of iso-octane/toluene mixture (Table 5 for P1–P5).

While, with Mechanism 2, it was over-predicted by around 100% for A3 and around 600% for A4 in SF flames, as described previously [3].

4. Concluding remarks

The new kinetic mechanism has been developed for gasoline surrogate fuels (n-heptane, iso-octane and toluene) to improve the model prediction of the concentrations of PAHs beyond ben-

zene and up to coronene. The density functional theory (DFT) and the transition state theory (TST) have been adopted to determine the reaction energetics and kinetics, respectively, for several PAH reactions. B3LYP/6-311++G(d,p) level of theory was used for all the quantum calculations.

To validate the newly developed mechanism, 6 premixed laminar flames of n-heptane, iso-octane, benzene and ethylene, and 44 counterflow diffusion flames of iso-octane/toluene and n-heptane/

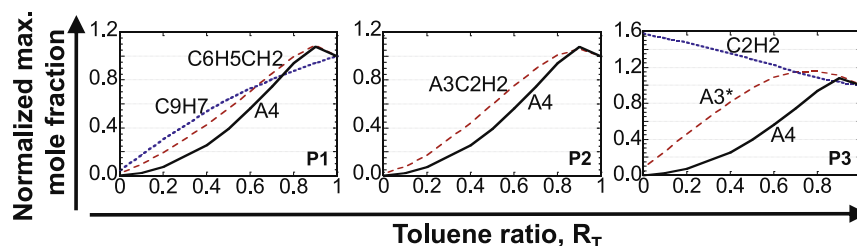


Fig. 14. Normalized maximum mole fractions of the species involved in the pathways responsible for the synergistic effect of A4 in the SFO flame of iso-octane/toluene mixture (Table 6 for P1–P5).

Table 6

Reactions leading to the synergistic behavior of pyrene (A4). Most of the reactions shown in the pathways are reversible and the arrows only show the direction of the progress of the reaction. The pathways indicate possible routes for the reactions to take place.

Pathway	Reactions
P1	$\text{C}_6\text{H}_5\text{CH}_2 + \text{C}_9\text{H}_7 \rightarrow \text{A}_4 + 2\text{H}_2$ Pathway for A4 formation through this route:
P2	$\text{A}_3\text{C}_2\text{H}_2^* \rightarrow \text{A}_4 + \text{H}$ Pathway for A3C2H2* formation:
P3	$\text{A}_3^* + \text{C}_2\text{H}_2 \rightarrow \text{A}_4 + \text{H}$

toluene mixtures have been simulated. The prediction of mole fraction profiles for large PAHs (A3 and A4) is significantly improved with the new mechanism in the n-heptane flames. In the other premixed laminar flames (for which PAH concentrations were available), the concentrations of large PAHs match reasonably well with their observed values. Those reactions containing odd number of carbon atoms such as C_3H_3 , C_5H_5 and C_9H_7 played a significant role in the formation of large PAHs.

The new mechanism was also able to predict the synergistic effect shown by PAHs in the counterflow diffusion flames of the n-heptane/toluene and iso-octane/toluene mixtures, which was not predicted well with two recently published mechanisms. The reactions that led to this effect in PAHs (A1–A4) were identified in the soot formation/oxidation (SFO) flames of iso-octane/toluene mixtures, and it was found that multiple reactions can be responsible for it. The profiles of the large PAHs (A4–A7) are in good agreement with the laser induced incandescence (LII) signals, which could imply the involvement of large PAHs in the formation of soot particles. Soot modeling efforts are required in the future to

account for the inception and growth of soot particles through PAHs larger than A4.

Acknowledgments

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Appendix A. Supplementary material

The comparison between the computed and the observed ignition delay times for several fuels (n-heptane, iso-octane, toluene and benzene) are shown in the supplementary material. The reaction mechanism file along with the thermodynamic and the transport data files used to generate the results shown in this paper are also provided in the CHEMKIN format. Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2011.08.011](https://doi.org/10.1016/j.combustflame.2011.08.011).

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